LAND REFORMS, GREEN REVOLUTION AND CROPPING SYSTEMS

LAND-REFORM INSTRUMENT MAP → LAW-TO-OUTCOME RAIL → STATE REFORM COMPARISON → GREEN-REVOLUTION PACKAGE → CONCENTRATION LEGACY → CROPPING METRICS FORK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 LAND-REFORM INSTRUMENT MAP

LAND-REFORM INSTRUMENT MAP | INTERMEDIARY ABOLITION | CULTIVATOR-STATE LINK | RENT + SECURITY + RECORDED CULTIVATOR | PERMISSIBLE HOLDING + SURPLUS LAND | WORKABLE PARCELS + ENFORCEABLE RIGHTS

INTERMEDIARY ABOLITION → cultivator-state link

TENANCY → rent + security + recorded cultivator → CEILINGS → permissible holding + surplus land → CONSOLIDATION / RECORDS → workable parcels + enforceable rights

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
• INTERMEDIARY ABOLITION → cultivator-state link • TENANCY → rent + security + recorded cultivator	• CEILINGS → permissible holding + surplus land • CONSOLIDATION / RECORDS → workable parcels + enforceable rights	• INTERMEDIARY ABOLITION → cultivator-state link • TENANCY → rent + security + recorded cultivator

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONSOLIDATION / RECORDS → workable parcels + enforceable rights.

01

STAGE 01 LAW-TO-OUTCOME RAIL

LAW-TO-OUTCOME RAIL | STATE LAW | ACCURATE RECORDS | LOCAL ADMINISTRATION + MOBILISATION | ACTUAL TENURE | REDISTRIBUTION OUTCOME

STATE LAW → ACCURATE RECORDS → LOCAL ADMINISTRATION + MOBILISATION → ACTUAL TENURE / REDISTRIBUTION OUTCOME

CONCEPT / ACCOUNTING BASIS
- STATE LAW
- ACCURATE RECORDS

TRANSMISSION / RECONCILIATION
- LOCAL ADMINISTRATION + MOBILISATION
- ACTUAL TENURE / REDISTRIBUTION OUTCOME

DISTRIBUTIONAL / STABILITY LIMIT
- STATE LAW
- ACCURATE RECORDS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LOCAL ADMINISTRATION + MOBILISATION; ACTUAL TENURE / REDISTRIBUTION OUTCOME.

02

STAGE 02 STATE REFORM COMPARISON

STATE REFORM COMPARISON | OPERATION BARGA | SHARECROPPER RECORDING | TENANCY ABOLITION + CEILING DEPTH | J&K 1950 | EARLY ESTATE RESTRUCTURING | COMPARE CONTEXT; DO NOT COPY OUTCOMES MECHANICALLY

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
OPERATION BARGA → sharecropper recording	J&K 1950 → early estate restructuring	OPERATION BARGA → sharecropper recording
KERALA → tenancy abolition + ceiling depth	COMPARE context; do not copy outcomes mechanically	KERALA → tenancy abolition + ceiling depth

MEASURE / INSTITUTION
- OPERATION BARGA → sharecropper recording
- KERALA → tenancy abolition + ceiling depth

DECISIVE DISTINCTION
- J&K 1950 → early estate restructuring
- COMPARE context; do not copy outcomes mechanically

VINTAGE / LEGAL LIMIT
- OPERATION BARGA → sharecropper recording
- KERALA → tenancy abolition + ceiling depth

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: J&K 1950 → early estate restructuring; COMPARE context; do not copy outcomes mechanically.

03

STAGE 03 GREEN-REVOLUTION PACKAGE

GREEN-REVOLUTION PACKAGE | HYV SEED + ASSURED WATER | FERTILISER + CREDIT + EXTENSION | PROCUREMENT + PRICE SUPPORT | PACKAGE EFFECT | YIELD + MARKETED SURPLUS

HYV SEED + ASSURED WATER → FERTILISER + CREDIT + EXTENSION → PROCUREMENT + PRICE SUPPORT → PACKAGE EFFECT → yield + marketed surplus

CONCEPT / ACCOUNTING BASIS
- HYV SEED + ASSURED WATER
- FERTILISER + CREDIT + EXTENSION

TRANSMISSION / RECONCILIATION
- PROCUREMENT + PRICE SUPPORT
- PACKAGE EFFECT → yield + marketed surplus

DISTRIBUTIONAL / STABILITY LIMIT
- HYV SEED + ASSURED WATER
- FERTILISER + CREDIT + EXTENSION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PROCUREMENT + PRICE SUPPORT; PACKAGE EFFECT → yield + marketed surplus.

04

STAGE 04 CONCENTRATION LEGACY

CONCENTRATION LEGACY | IRRIGATED NORTHWEST | RICE SPECIALISATION | FOODGRAIN SURPLUS | GROUNDWATER + SOIL + REGIONAL STRESS

IRRIGATED NORTHWEST → WHEAT / RICE SPECIALISATION → FOODGRAIN SURPLUS → GROUNDWATER + SOIL + REGIONAL STRESS

CONCEPT / ACCOUNTING BASIS
- IRRIGATED NORTHWEST
- WHEAT / RICE SPECIALISATION

TRANSMISSION / RECONCILIATION
- FOODGRAIN SURPLUS
- GROUNDWATER + SOIL + REGIONAL STRESS

DISTRIBUTIONAL / STABILITY LIMIT
- IRRIGATED NORTHWEST
- WHEAT / RICE SPECIALISATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FOODGRAIN SURPLUS; GROUNDWATER + SOIL + REGIONAL STRESS.

05

STAGE 05 CROPPING METRICS FORK

CROPPING METRICS FORK | CROP-AREA DISTRIBUTION | GROSS CROPPED | NET SOWN AREA | ALLIED SHIFT | EACH NEEDS STATED TIME + GEOGRAPHY

PATTERN → crop-area distribution

INTENSITY → gross cropped / net sown area → DIVERSIFICATION → crop / allied shift → EACH needs stated time + geography

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
• PATTERN → crop-area distribution • INTENSITY → gross cropped / net sown area	• DIVERSIFICATION → crop / allied shift • EACH needs stated time + geography	• PATTERN → crop-area distribution • INTENSITY → gross cropped / net sown area

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTENSITY → gross cropped / net sown area.

06

STAGE 06 DIVERSIFICATION DECISION BOARD

DIVERSIFICATION DECISION BOARD | AGRO-CLIMATE + WATER | EXPECTED RETURN + VOLATILITY | CREDIT + INSURANCE + EXTENSION | STORAGE + PROCESSING + BUYER ACCESS

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
AGRO-CLIMATE + WATER	CREDIT + INSURANCE + EXTENSION	AGRO-CLIMATE + WATER
EXPECTED RETURN + VOLATILITY	STORAGE + PROCESSING + BUYER ACCESS	EXPECTED RETURN + VOLATILITY

MEASURE / INSTITUTION
- AGRO-CLIMATE + WATER
- EXPECTED RETURN + VOLATILITY

DECISIVE DISTINCTION
- CREDIT + INSURANCE + EXTENSION
- STORAGE + PROCESSING + BUYER ACCESS

VINTAGE / LEGAL LIMIT
- AGRO-CLIMATE + WATER
- EXPECTED RETURN + VOLATILITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CREDIT + INSURANCE + EXTENSION; STORAGE + PROCESSING + BUYER ACCESS.

07

STAGE 07 MILLET TRANSITION RAIL

MILLET TRANSITION RAIL | DRYLAND SUITABILITY | LOWER WATER EXPOSURE | RESILIENCE VALUE | NEED PROCESSING + PROCUREMENT + DEMAND

DRYLAND SUITABILITY → LOWER WATER EXPOSURE → NUTRITION / RESILIENCE VALUE → NEED processing + procurement + demand

CONCEPT / ACCOUNTING BASIS
- DRYLAND SUITABILITY
- LOWER WATER EXPOSURE

TRANSMISSION / RECONCILIATION
- NUTRITION / RESILIENCE VALUE
- NEED processing + procurement + demand

DISTRIBUTIONAL / STABILITY LIMIT
- DRYLAND SUITABILITY
- LOWER WATER EXPOSURE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NUTRITION / RESILIENCE VALUE; NEED processing + procurement + demand.

08

STAGE 08 WATER-CROP LOCK-IN

WATER-CROP LOCK-IN | PROCUREMENT + CHEAP PUMPING | WATER-INTENSIVE CROP CHOICE | SOIL STRESS | DIVERSIFICATION BECOMES HARDER

CONCEPT / ACCOUNTING BASIS
- PROCUREMENT + CHEAP PUMPING
- WATER-INTENSIVE CROP CHOICE

TRANSMISSION / RECONCILIATION
- AQUIFER / SOIL STRESS
- DIVERSIFICATION BECOMES HARDER

DISTRIBUTIONAL / STABILITY LIMIT
- PROCUREMENT + CHEAP PUMPING
- WATER-INTENSIVE CROP CHOICE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER-INTENSIVE CROP CHOICE; AQUIFER / SOIL STRESS; DIVERSIFICATION BECOMES HARDER.

09

STAGE 09 SCIENTIST-ENGINEER DISTINCTION

SCIENTIST-ENGINEER DISTINCTION | SLUICE GATES + BLOCK IRRIGATION | SEMI-DWARF CROP ADAPTATION | ASSURED WATER SYSTEMS | CROP SCIENCE | YIELD RESPONSE

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
VISVESVARAYA → sluice gates + block irrigation	ENGINEERING → assured water systems	VISVESVARAYA → sluice gates + block irrigation
SWAMINATHAN → semi-dwarf crop adaptation	CROP SCIENCE → yield response	SWAMINATHAN → semi-dwarf crop adaptation

MEASURE / INSTITUTION
- VISVESVARAYA → sluice gates + block irrigation
- SWAMINATHAN → semi-dwarf crop adaptation

DECISIVE DISTINCTION
- ENGINEERING → assured water systems
- CROP SCIENCE → yield response

VINTAGE / LEGAL LIMIT
- VISVESVARAYA → sluice gates + block irrigation
- SWAMINATHAN → semi-dwarf crop adaptation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENGINEERING → assured water systems; CROP SCIENCE → yield response.

10

STAGE 10 AGRARIAN TRADE-OFFS

AGRARIAN TRADE-OFFS | SECURE TENURE | OWNER CONCERNS | VULNERABLE-USER SAFEGUARDS | HIGH YIELD | FOOD SECURITY | ECOLOGICAL COST | MARKET RISK

SECURE TENURE → investment / owner concerns

CONSOLIDATION → scale / vulnerable-user safeguards → HIGH YIELD → food security / ecological cost → DIVERSIFICATION → resilience / market risk

CONCEPT / ACCOUNTING BASIS
- SECURE TENURE → investment / owner concerns
- CONSOLIDATION → scale / vulnerable-user safeguards

TRANSMISSION / RECONCILIATION
- HIGH YIELD → food security / ecological cost
- DIVERSIFICATION → resilience / market risk

DISTRIBUTIONAL / STABILITY LIMIT
- SECURE TENURE → investment / owner concerns
- CONSOLIDATION → scale / vulnerable-user safeguards

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGH YIELD → food security / ecological cost.

11

STAGE 11 AGRARIAN ANSWER SPINE

AGRARIAN ANSWER SPINE | DEFINE RIGHTS, RECORDS AND HOLDING STRUCTURE | TRACE TECHNOLOGY + WATER + MARKET PACKAGE | COMPARE REGIONAL OUTCOMES AND TRADE-OFFS | CONCLUDE WITH VIABLE DIVERSIFICATION

DEFINE rights, records and holding structure → TRACE technology + water + market package → COMPARE regional outcomes and trade-offs → CONCLUDE with viable diversification

CONCEPT / ACCOUNTING BASIS
- DEFINE rights, records and holding structure
- TRACE technology + water + market package

TRANSMISSION / RECONCILIATION
- COMPARE regional outcomes and trade-offs
- CONCLUDE with viable diversification

DISTRIBUTIONAL / STABILITY LIMIT
- DEFINE rights, records and holding structure
- TRACE technology + water + market package

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMPARE regional outcomes and trade-offs; CONCLUDE with viable diversification.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | SECURE TENURE CAN IMPROVE INVESTMENT INCENTIVES, YET POORLY DESIGNED | LEASING REFORM CAN SEPARATE OWNERSHIP FROM EFFICIENT CULTIVATION WHILE | LAND RECORDS ARE FISCAL, CREDIT, PLANNING AND DISPUTE-RESOLUTION INFRASTRUCTURE | GREEN REVOLUTION GAINS CREATED FOOD SECURITY BUT ALSO PATH | CROP DIVERSIFICATION REQUIRES DEMAND, PROCESSING, INSURANCE AND LOGISTICS, NOT | ALLIED-SECTOR GROWTH CAN STABILISE INCOME BECAUSE PRODUCTION CYCLES AND | STATE REVENUE AND LAND DEPARTMENTS

OPTIONAL MODEL / MEASUREMENT DEPTH
- CAUTION: Secure tenure can improve investment incentives, yet poorly designed ceilings may
- CAUTION: Leasing reform can separate ownership from efficient cultivation while protecting
- CAUTION: Land records are fiscal, credit, planning and dispute-resolution infrastructure, not
- CAUTION: Green Revolution gains created food security but also path dependence in water,

VINTAGE, BASKET OR LEGAL LIMIT
- CAUTION: Crop diversification requires demand, processing, insurance and logistics, not price
- CAUTION: Allied-sector growth can stabilise income because production cycles and risks differ
- State revenue and land departments: implement land records, tenancy and ceiling
- Agricultural universities and extension systems: adapt seed and agronomy to local

COMPARATIVE POLICY NUANCE
- Procurement and irrigation agencies: shape crop profitability and regional
- Panchayats, co-operatives and farmer organisations: support local records, input
- CAUTION: Operation Barga is used to illustrate how political mobilisation and tenancy recording
- CAUTION: Punjab-Haryana cereal specialisation shows how irrigation and procurement can lock a

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels